Bernhard Riemann — Partielle Differentialgleichungen und deren Anwendung auf physikalische Fragen (1869)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 14 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Bernhard Riemann, who died at thirty-nine and left half his legacy as paper in other people's hands. These lectures on partial differential equations and their application to physical questions were delivered at Göttingen and preserved by his student Karl Hattendorff, who brought them into print in 1869, three years after Riemann's death. Darboux wrote in that same year that the little volume was so clear it could be put straight into students' hands. English students have never been handed it. This presentation is dedicated to Riemann, and to the pupil who set his lectures down, in the hope that a book praised from the first for its clarity may one day teach in English as it has long taught in German.

The story

Georg Friedrich Bernhard Riemann (1826–1866) died at thirty-nine, and half of his legacy is posthumous paper preserved by his students. Among that paper are his Göttingen lectures on partial differential equations and their application to physical questions — delivered in 1854–55, 1860–61, and 1862, and brought into print by his student Karl Hattendorff in 1869. A partial differential equation describes how a quantity changes in several directions at once, and Riemann's

lectures range across the great physical subjects the equations govern: acoustics, heat, hydrodynamics, and electricity, including Riemann's own electrodynamics, in which he let the deformation of the ether carry the force — the curvature idea in embryo. The lectures were admired at once for their clarity. Darboux wrote to Houël in 1869 that the little volume was so clear it could be put straight into students' hands.

English students, however, have never been handed it. Riemann in English is a patchwork — the famous Habilitation lecture in Clifford's translation of 1873, and scattered papers — and a French translation of the collected works appeared as early as 1898. The lectures were never part of that. They have lived, for more than a century and a half, as a German-reading classic.

The work and what survives

The edition history needs care, because the book changed as it was reissued. The editions run: 1869 (Hattendorff), second and third (the third in 1882), then a fourth (1900–01) "revised and rewritten" by Heinrich Weber — that is, the later book is increasingly Weber's, not Riemann's. The rescue target in this research is the 1869 Hattendorff text, about 332 pages: Riemann's lectures as his pupil set them down. German scans circulate openly, and the 1869 text is the basis of modern reprints; the Internet Archive identifier is to be pinned at build time. The work is in the public domain: an 1869 imprint; the author died in 1866 and the editor in 1903.

A reviewer's check confirms the target and sharpens its boundary. Bibliographies list only the German editions — 1869, 1876, and 1882, with HathiTrust page images for all three, per the Online Books Page — and their Weber and Frank–Mises descendants. The reviewer also confirms, from a contemporary *Nature* review, that the Weber 1900–01 edition is a rewriting.

Its English fortunes

One neighbouring fact must be stated plainly, so that nothing here overclaims. A complete-ish English *Collected Papers* does exist, in copyright: Kendrick Press, 2004, translated by Baker, Christenson, and Orde, from the 1892 edition. But it renders the papers in Weber's *Gesammelte Werke* volume — omitting, per the MAA review, an article on the mechanism of the ear and two philosophy fragments — and the PDE lectures are a separate book, not in that volume. This project must therefore never be presented as "Riemann's works in English."

Kendrick exists. The honest claim concerns the lectures specifically.

The second-pass hardening check called for in this research has been run and discharged, across the Online Books Page (the Hattendorff author record), the DSB bibliography, the *Nature* 1900 review, the modern print-on-demand reprint graph, and the Kendrick and MAA record. Its result: **no English edition of the PDE lectures exists anywhere in the edition graph.** Every edition recorded is German (Vieweg 1869, 1876, 1882); the Weber and Frank–Mises descendants are revisions, not translations; every modern reprint is explicitly catalogued "(German Edition)"; and the only English Riemann volumes are of the *Gesammelte Werke*, which excludes the lectures. The wording "**No English translation of the lectures found after research**" now stands census-backed.

What completing it would take

Completing the work would mean a translation of the 1869 German, with the equations re-set in modern notation — one continuous volume of about 330 pages. The review standard set for it is a working mathematician in partial differential equations, together with a reader of period German.

Sources

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